ARC-AGI Milestone C Journal
CS7637 Knowledge-Based AI

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1 INTRODUCTION

In this milestone, I focused on expanding my agent's ability to handle a wider variety of ARC problems by introducing additional transformation rules and improving rule validation. The agent still follows a rule-based approach. For each problem, it evaluates a list of candidate rules against all training examples and selects a rule only if it perfectly explains the entire set. Once a rule is selected, it is applied directly to the test input. Through this process, the agent successfully solved **6 problems** in Set C. Some of these were handled by previously implemented rules, while others required new rules developed during this milestone.

2 AGENT DESIGN AND FUNCTIONING

he agent operates using a strict validation process:

1. Iterate through candidate rules
2. Apply each rule to all training inputs
3. Reject the rule if any mismatch occurs
4. Accept the rule only if all examples match
5. Apply the rule to the test input

This guarantees that the selected rule generalizes across the full training set.

During this milestone, I introduced new rules based on observed patterns:

- Region-based recoloring using corner markers
- Diagonal propagation
- Symmetry-based reconstruction
- Frequency-based reconstruction
- Pattern comparison between split regions

3 SOLVED PROBLEMS AND RULES

3.1 Problem 3428a4f5

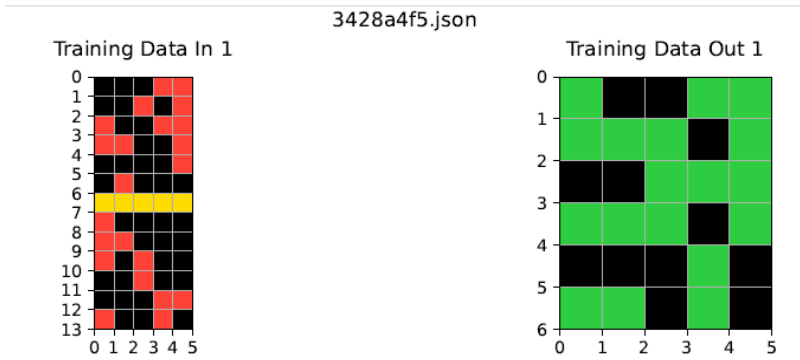

Figure 1—Pattern Comparison Between Two Halves

Rule Description: Top vs Bottom Region Comparison (Difference Mask)

- The grid is split horizontally using the row of 4s as a separator
- The top region and bottom region are compared cell-by-cell
- If the two corresponding cells are different → output 3
- If they are the same → output 0

This is essentially a **difference mask between two panels**, similar to an XOR-like comparison but applied spatially.

3.2 Problem 3de23699

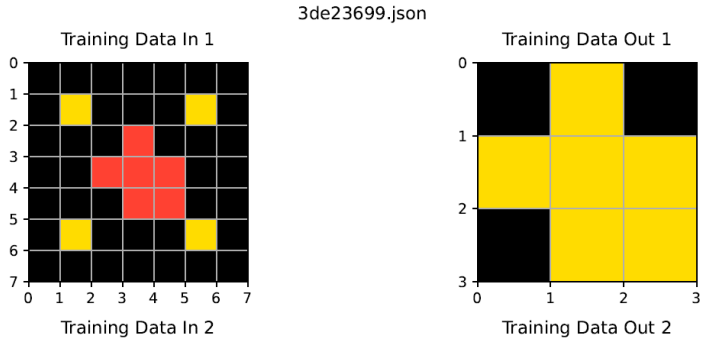

Figure 2—Corner-Based Region Recoloring and Cropping

Rule Description: Corner-based Region Recolor + Crop

- Identify four non-zero corner markers

- Use them as boundaries of a region
- Replace all non-zero values inside with the marker color
- Crop the region

3.3 Problem 5c0a986e

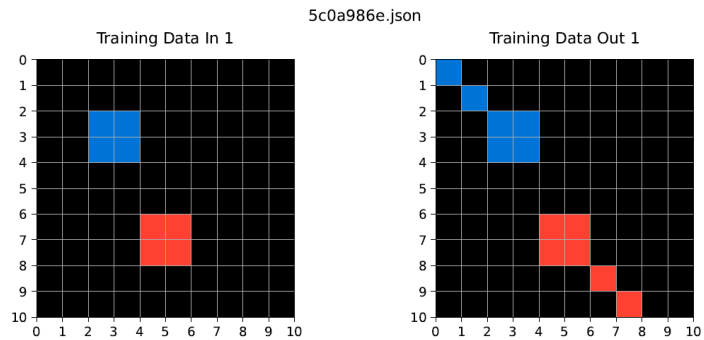

Figure 3—Directional Diagonal Propagation

Rule Description: Diagonal Expansion

- Value 1 propagates diagonally toward top-left (−1, −1)
- Value 2 propagates diagonally toward bottom-right (+1, +1)

3.4 Problem 62c24649

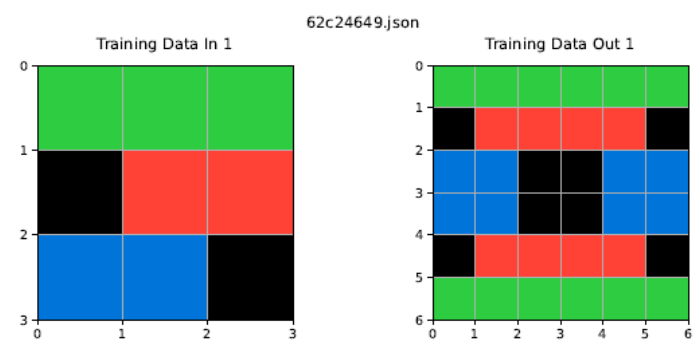

Figure 4—Symmetry-Based Reconstruction

Rule Description: Multi-Axis Symmetry

- Reflect across vertical axis
- Reflect across horizontal axis
- Combine all reflections

3.5 Problem 74dd1130

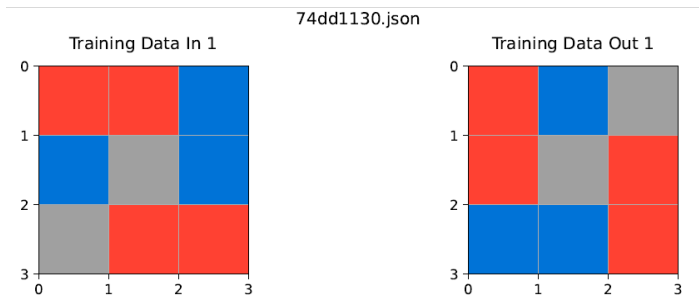

Figure 5—Rotation and Reflection Transformation

Rule Description: Rotate 90° Counterclockwise + Vertical Flip

- First rotate the grid 90 degrees
- Then flip across the vertical axis

Equivalent interpretation:

- Columns of the input become rows in the output

This transformation rearranges directional structure into aligned rows.

3.6 Problem 9af7a82c

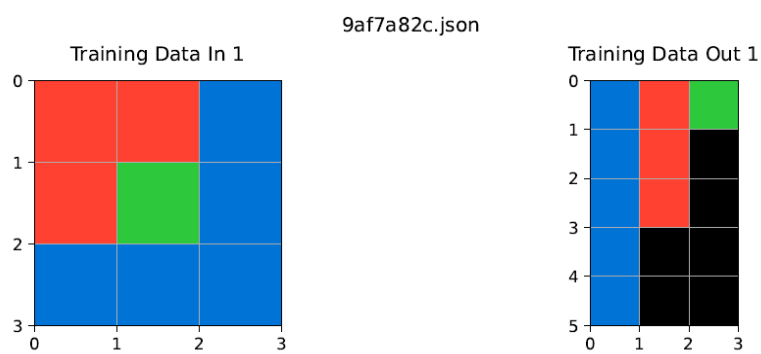

Figure 6—Frequency-Based Column Construction

Rule Description: Frequency-Based Reconstruction

- Count occurrences of each value
- Sort by frequency (descending)
- Use highest frequency as output height

- Fill columns vertically according to counts

4 PERFORMANCE, STRENGTHS, AND LIMITATIONS

The agent successfully solved six problems in this milestone: 3428a4f5, 3de23699, 5c0a986e, 62c24649, 74dd1130, and 9af7a82c. Among these, three were handled using rules that were already implemented earlier, while the remaining three required new rules developed during this milestone. This shows that expanding the rule set directly improved the agent's ability to cover a wider range of patterns.

The agent performs well when the transformation is consistent across all training examples and can be expressed as a single, clearly defined rule. In particular, it handles structured patterns such as symmetry, diagonal propagation, and frequency-based reconstruction effectively. These types of problems tend to have strong visual or positional regularities, which makes them easier to capture using deterministic rules.

However, the agent still struggles in cases where no single rule fully explains all training examples or when multiple transformations need to be applied in sequence. Because the agent strictly requires a rule to match every training example exactly, even a partially correct rule is rejected. This makes the system less flexible and limits its ability to generalize in more complex situations.

From an efficiency standpoint, each rule operates by scanning the grid, resulting in a time complexity of approximately $O(n^2)$ per rule. Since the grid sizes in ARC are relatively small and the number of candidate rules is limited, the overall runtime remains manageable.

5 CONCLUSION

In this milestone, I expanded the agent by introducing several new transformation rules and refining the existing rule selection process. As a result, the agent was able to solve six problems from Set C, including cases involving symmetry, diagonal propagation, region-based reasoning, and frequency-based reconstruction. While the current approach works well for structured problems, further improvements are needed to handle more complex and compositional transformations.